

Theory of Computation

DPDA Design and Implementation

A decorative teal underline consisting of two parallel curved lines that taper at both ends, positioned below the main title.

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PROBLEM

- A deterministic pushdown automaton has at most one transition for the same combination of input symbol, state, and top stack symbol
- Has to recognize context free language $L = a^n b^n, n \geq 0$
- Deterministic, one step look ahead
- Equal number of a's and b's

DPDA



STATES

$p, q, qa, qb, q\$$

RULES

$S \rightarrow aSb$

$S \rightarrow e$

TRANSITION RULES

(1) $(p, e, e) \rightarrow (q, S)$

(2) $(q, a, e) \rightarrow (qa, e)$

(3) $(qa, e, a) \rightarrow (q, e)$

(4) $(q, b, e) \rightarrow (qb, e)$

(5) $(qb, e, b) \rightarrow (q, e)$

(6) $(q, \$, e) \rightarrow (q\$, e)$

(7) $(qa, e, S) \rightarrow (qa, aSb)$

(8) $(qb, e, S) \rightarrow (qb, e)$

MY DPDA

```

1 #include <iostream>
2 #include <string>
3 #include <stack>
4 #include <vector>
5 #include <iomanip>
6
7 using namespace std;
8
9
10 class DPDA{
11
12     private:
13         string state;
14         stack<string> DPDAstack;
15         string input;
16         int pos;
17         int stepNum;
18
19     public:
20         DPDA(string stringInput){
21             input = stringInput;
22             state = "p";
23             pos = 0;
24             stepNum = 0;
25         }
26
27         void printStep(string delta, string rule) {
28             string top;
29             if (DPDAstack.empty()){
30                 top = "e";
31             }
32             else{
33                 top = DPDAstack.top();
34             }
35
36             string unread;
37             if (pos >= (int)input.size()){
38                 unread = "e";
39             }
40             else {
41                 unread = input.substr(pos);
42             }
43             cout << left

```

```

43             cout << left
44                 << setw(8) << stepNum
45                 << setw(8) << state
46                 << setw(15) << unread
47                 << setw(15) << getStackContents()
48                 << setw(15) << delta
49                 << setw(10) << rule << endl;
50         }
51
52         string getStackContents() {
53             if (DPDAstack.empty()){
54                 return "e";
55             }
56
57             stack<string> temp = DPDAstack; // copy the stack
58             string result = "";
59
60             while (!temp.empty()) {
61                 result += temp.top(); // add top to result
62                 temp.pop(); // remove top
63             }
64             return result;
65         }
66
67         void run(){
68
69             cout << left
70                 << setw(8) << "Step"
71                 << setw(8) << "State"
72                 << setw(15) << "Unread Input"
73                 << setw(15) << "Stack"
74                 << setw(15) << "Delta Used"
75                 << setw(10) << "Rule Used" << endl;
76
77             printStep("", "");
78             while (state != "q$") {
79
80                 string delta = "";
81                 string rule = "";
82
83                 char currentInput = input[pos]; //what is read

```

```

string stackTop = "";
if (!DPDAstack.empty()) {
    stackTop = DPDAstack.top(); //top of stack
}

//rule 1 (p, e, e) -> (q, S)
if (state == "p") {
    delta = "11";
    state = "q";
    DPDAstack.push("S");
}

//rule 2 (q, a, e) -> (qa, e)
else if (state == "q" && currentInput == 'a') {
    delta = "22";
    state = "qa";
    pos++;
}

//rule 3 (qa, e, a) -> (q, e)
else if (state == "qa" && stackTop == "a") {
    delta = "33";
    state = "q";
    DPDAstack.pop();
}

//rule 4 (q, b, e) -> (qb, e)
else if (state == "q" && currentInput == 'b') {
    delta = "44";
    state = "qb";
    pos++;
}

//rule 5 (qb, e, b) -> (q, e)
else if (state == "qb" && stackTop == "b") {
    delta = "55";
    state = "q";
    DPDAstack.pop();
}

```

```

}

//rule 6 (q, $, e) -> (q$, e)
else if (state == "q" && currentInput == '$') {
    delta = "66";
    state = "q$";
    pos++;
}

//rule 7 (qa, e, S) -> (qa, aSb)
else if (state == "qa" && stackTop == "S") {
    delta = "77";
    rule = "S -> aSb";
    DPDAstack.pop();
    DPDAstack.push("b");
    DPDAstack.push("S");
    DPDAstack.push("a");
}

//rule 8 (qb, e, S) -> (qb, e)
else if (state == "qb" && stackTop == "S") {
    rule = "S -> e";
    delta = "88";
    DPDAstack.pop();
}

stepNum++;
printStep(delta, rule);
}

if (state == "q$") {
    cout << "ACCEPTED" << endl;
}
else {
    cout << "REJECTED" << endl;
}
}

```

Testcase

ab\$

Input: ab\$

Step	State	Unread Input	Stack	Delta Used	Rule Used
0	p	ab\$	e		
1	q	ab\$	S	1	
2	qa	b\$	S	2	
3	qa	b\$	aSb	7	$S \rightarrow aSb$
4	q	b\$	Sb	3	
5	qb	\$	Sb	4	
6	qb	\$	b	8	$S \rightarrow e$
7	q	\$	e	5	
8	q\$	e	e	6	

ACCEPTED

Inputs

$a^2b^2\$$

Input: aabb\$							
Step	State	Unread	Input	Stack	Delta	Used	Rule Used
0	p	aabb\$		e			
1	q	aabb\$		S	1		
2	qa	abb\$		S	2		
3	qa	abb\$		aSb	7		S → aSb
4	q	abb\$		Sb	3		
5	qa	bb\$		Sb	2		
6	qa	bb\$		aSbb	7		S → aSb
7	q	bb\$		Sbb	3		
8	qb	b\$		Sbb	4		
9	qb	b\$		bb	8		S → e
10	q	b\$		b	5		
11	qb	\$		b	4		
12	q	\$		e	5		
13	q\$	e		e	6		
ACCEPTED							

$a^3b^3\$$

Input: aaabbb\$							
Step	State	Unread	Input	Stack	Delta	Used	Rule Used
0	p	aaabbb\$		e			
1	q	aaabbb\$		S	1		
2	qa	aabbb\$		S	2		
3	qa	aabbb\$		aSb	7		S → aSb
4	q	aabbb\$		Sb	3		
5	qa	abbb\$		Sb	2		
6	qa	abbb\$		aSbb	7		S → aSb
7	q	abbb\$		Sbb	3		
8	qa	bbb\$		Sbb	2		
9	qa	bbb\$		aSbbb	7		S → aSb
10	q	bbb\$		Sbbb	3		
11	qb	bb\$		Sbbb	4		
12	qb	bb\$		bbb	8		S → e
13	q	bb\$		bb	5		
14	qb	b\$		bb	4		
15	q	b\$		b	5		
16	qb	\$		b	4		
17	q	\$		e	5		
18	q\$	e		e	6		
ACCEPTED							

Inputs

$a^4b^4\$$

Input: aaaabbbb\$

Step	State	Unread Input	Stack	Delta	Used	Rule Used
0	p	aaaabbbb\$	e			
1	q	aaaabbbb\$	S	1		
2	qa	aaabbbb\$	S	2		
3	qa	aaabbbb\$	aSb	7		S → aSb
4	q	aaabbbb\$	Sb	3		
5	qa	aabbbb\$	Sb	2		
6	qa	aabbbb\$	aSbb	7		S → aSb
7	q	aabbbb\$	Sbb	3		
8	qa	abbbb\$	Sbb	2		
9	qa	abbbb\$	aSbbb	7		S → aSb
10	q	abbbb\$	Sbbb	3		
11	qa	bbbb\$	Sbbb	2		
12	qa	bbbb\$	aSbbb	7		S → aSb
13	q	bbbb\$	Sbbb	3		
14	qb	bbb\$	Sbbb	4		
15	qb	bbb\$	bbbb	8		S → e
16	q	bbb\$	bbb	5		
17	qb	bb\$	bbb	4		
18	q	bb\$	bb	5		
19	qb	b\$	bb	4		
20	q	b\$	b	5		
21	qb	\$	b	4		
22	q	\$	e	5		
23	q\$	e	e	6		

ACCEPTED

$a^5b^5\$$

Input: aaaaabbbb\$

Step	State	Unread Input	Stack	Delta	Used	Rule Used
0	p	aaaaabbbb\$	e			
1	q	aaaaabbbb\$	S	1		
2	qa	aaaabbbb\$	S	2		
3	qa	aaaabbbb\$	aSb	7		S → aSb
4	q	aaaabbbb\$	Sb	3		
5	qa	aaabbbb\$	Sb	2		
6	qa	aaabbbb\$	aSbb	7		S → aSb
7	q	aaabbbb\$	Sbb	3		
8	qa	aabbbb\$	Sbb	2		
9	qa	aabbbb\$	aSbbb	7		S → aSb
10	q	aabbbb\$	Sbbb	3		
11	qa	abbbb\$	Sbbb	2		
12	qa	abbbb\$	aSbbb	7		S → aSb
13	q	abbbb\$	Sbbb	3		
14	qa	bbbb\$	Sbbb	2		
15	qa	bbbb\$	aSbbbb	7		S → aSb
16	q	bbbb\$	Sbbbb	3		
17	qb	bbb\$	Sbbbb	4		
18	qb	bbb\$	bbbb	8		S → e
19	q	bbb\$	bbbb	5		
20	qb	bbb\$	bbbb	4		
21	q	bbb\$	bbb	5		
22	qb	bb\$	bbb	4		
23	q	bb\$	bb	5		
24	qb	b\$	bb	4		
25	q	b\$	b	5		
26	qb	\$	b	4		
27	q	\$	e	5		
28	q\$	e	e	6		

ACCEPTED

Inputs

$a^6b^6\$$

Input: aaaaaabbbbb\$						
Step	State	Unread Input	Stack	Delta	Used	Rule Used
0	p	aaaaaabbbbb\$	e			
1	q	aaaaaabbbbb\$	S	1		
2	qa	aaaaaabbbbb\$	S	2		
3	qa	aaaaaabbbbb\$	aSb	7		S → aSb
4	q	aaaaaabbbbb\$	Sb	3		
5	qa	aaaabbbbb\$	Sb	2		
6	qa	aaaabbbbb\$	aSbb	7		S → aSb
7	q	aaaabbbbb\$	Sbb	3		
8	qa	aaabbbbb\$	Sbb	2		
9	qa	aaabbbbb\$	aSbbb	7		S → aSb
10	q	aaabbbbb\$	Sbbb	3		
11	qa	aabbbbb\$	Sbbb	2		
12	qa	aabbbbb\$	aSbbbb	7		S → aSb
13	q	aabbbbb\$	Sbbbb	3		
14	qa	abbbbb\$	Sbbbb	2		
15	qa	abbbbb\$	aSbbbb	7		S → aSb
16	q	abbbbb\$	Sbbbb	3		
17	qa	bbbbb\$	Sbbbb	2		
18	qa	bbbbb\$	aSbbbb	7		S → aSb
19	q	bbbbb\$	Sbbbb	3		
20	qb	bbbbb\$	Sbbbb	4		
21	qb	bbbbb\$	bbbbb	8		S → e
22	q	bbbbb\$	bbbbb	5		
23	qb	bbbb\$	bbbbb	4		
24	q	bbbb\$	bbbb	5		
25	qb	bbb\$	bbbb	4		
26	q	bbb\$	bbb	5		
27	qb	bb\$	bbb	4		
28	q	bb\$	bb	5		
29	qb	b\$	bb	4		
30	q	b\$	b	5		
31	qb	\$	b	4		
32	q	\$	e	5		
33	q\$	e	e	6		

ACCEPTED



Thank you!